Fuel Low Pressure Indicating System – General Data

1. General

Fuel low pressure is indicated by caution messages displayed on the flight display panel, operated by fuel pressure switches mounted beneath the forward and aft fuselage tanks. On aircraft with wing tanks fitted (Mod S.O.O. 6247), additional pressure switches are provided to indicate low fuel pressure conditions in the wing tanks fuel supply lines to the engines.

A. Fuselage Fuel Low Pressure Indicating System

Low fuel pressure in the fuselage tanks is indicated by one of two methods. The first method is the Pmp1 and Pmp2 amber blocks displayed on fuel pane. These amber blocks will disappear during normal system operation, with Pmp2 being suppressed by the pump changeover control box when pump 1 is on and producing pressure. If pump 1 fails, then Pmp1 will become active and Pmp2 will remain off. The second method is through pump pressure CAS messages of Pump 1 Forward Press, Pump 1 Aft Press, Pump 2 Forward Press, and Pump 2 Aft Press. These messages are activated based on switch input that defines which pumps are being commanded on, and the associated pump pressure switch state. Each amber block and CAS message is operated seperately by a fuel pressure switch mounted below either No. 4 cell sump plate (forward tank) or below No. 5 cell sump plate (aft tank). Refer to PSM 1-64-2W for details.

B. Wing Fuel Low Pressure (PUMP FAIL) Indicating System
(Mod S.O.O. 6247)

Indication of pump failure or low fuel pressure in the engine fuel feed lines from the wing tanks is accomplished by caution messages, which can be operated when the applicable ENGINE (L or R) selection is made. Each Left Wing Fuel Pump and Right Wing Fuel Pump message (and the applicable amber fuel pane blocks) is operated by a pressure switch mounted in a four way fitting on the outboard side of the applicable wing tank, downstream from the wing tank fuel pressure pump. Incorporated in each pressure switch mounting is a bleed hole into the tank, which allows the pump to prime if it becomes vapor locked.

Refer to PSM 1-64-2W for electrical schematic for the Wing Fuel Low Pressure Indicating System.

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